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A-Dying-Pig

Yiran Lei

Education

2023 – Present — **Carnegie Mellon University**, Computer Science Department
Ph.D. in Computer Science (AI Networking & GPU communication)
Advisor: *Prof. Justine Sherry*

2020 – 2023 — **Tsinghua University**, Department of Computer Science and Technology
M.E. in Computer Science, GPA 3.76 / 4.0
Advisor: *Prof. Mingwei Xu*.

2016 – 2020 — **Tsinghua University**, School of Software
B.E. in Software Engineering, GPA 3.65 / 4.0
Thesis title: *Track Multiple Objects across Different Points of Views*.

Industry Internship

2025.5 – 2025.8 — **Microsoft Research**
Research intern, advisor: *Michael Papamichael*
Develop programmable congestion control algorithms for large-scale AI networks

2024.5 – 2025.2 — **MangoBoost Inc.**
Technical intern, advisor: *Eriko Nurvitadhi*
Optimize `AlltoAllv` communication for MoE workloads in modern GPU clusters

Research Internship

2022.9 – 2023.5 — **University of Washington**, Paul G. Allen School of Computer Science
Visiting student intern, advisor: *Prof. Arvind Krishnamurthy*
Build fast, consistent shared states for efficiently deploying network functions on both programmable switch and SmartNICs.

2021.9 – 2022.8 — **University of Pennsylvania**, Distributed Systems Lab
Remote intern, advisor: *Prof. Vincent Liu*
Reasonable about queue content at switches for understanding and debugging congestion.

2019.6 – 2019.8 — **UCLA**, Internet Research Lab
Visiting student intern, advisor: *Prof. Lixia Zhang*
Build NDN home IoT system.

Research Publications

1. **Yiran Lei**, Francisco Pereira, Arvind Krishnamurthy, and Justine Sherry. 2025. Switchnic: an hybrid architecture for network functions with fast and consistent shared state. 3, CoNEXT4, Article 46, (November 2025), 21 pages. DOI: `10.1145/3768993`. `https://doi.org/10.1145/3768993`.

2. **Yiran Lei**, Liangcheng Yu, Vincent Liu, and Mingwei Xu. 2022. Printqueue: performance diagnosis via queue measurement in the data plane. In *ACM SIGCOMM 2022 Conference (SIGCOMM '22), August 22–26, 2022, Amsterdam, Netherlands*, ACM, New York, NY, USA, 14 pages. DOI: `10.1145/3544216.3544257`.

3 **Yiran Lei**, Yu Zhou, Yunsenxiao Lin, Mingwei Xu, and Yangyang Wang. 2021. Dove: diagnosis-driven slo violation detection. In *2021 IEEE 29th International Conference on Network Protocols (ICNP)*, 1–11. DOI: 10.1109/ICNP52444.2021.9651986.

Teaching Assistant

2025.9 – 2025.12 — *Computer Networks and the Internet* (15-441/641), Justine Sherry, Carnegie Mellon University.
Give lectures about HTTP and data center networks; lead recitations.

2021.9 – 2022.1 — *The Principle of Computer Network* (40240513), Mingwei Xu, Tsinghua University.
Give lectures about IPv6.

Skills

Networking	NVSHMEM, RCCL/NCCL, DPDK, Mininet, Programmable Switch (Tofino), SmartNIC (Stingray, Bluefield-3 DOCA)
Math	Stochastic Process, Combinatorics, Calculus, Linear Algebra, Algorithms
Languages	English: TOEFL iBT 112 (30L, 29R, 25S, 28W), Chinese
Coding	P4, Python, C/C++, Javascript, Java, Assembly Language, SQL, Vitis HLS, CUDA/HIP
Other Systems	NVIDIA H200 GPU, AMD MI300X GPU, Linux Kernel, Raspberry PI, Arduino, TinyOS
Web Dev	Django, Vue.js, HTML5, Flask

Awards

2022	China National Scholarship
2021	Fellowship for Comprehensive Excellence (Second Class), Tsinghua University
2018	Second Award in Contemporary Undergraduate Mathematical Contest in Modeling, China
	Honorable Mention in Mathematical Contest in Modeling, USA
2017	Scholarship for Excellence in Study, Tsinghua University

Projects

2021 — **System Calls**, implementing fork, exec, spawn, link, user shell on *ucore* OS
C based. Grasp linux kernel and user space, file system, trap, system calls.

2019 — Reproduce the result of "*Deferred Neural Rendering: Image Synthesis using Neural Textures*"
OpenGL and UNet based. Implement multiple lighting models, e.g., Blinn-Phong and physical lighting model.

LowSQL Database, a high performance SQL database
Java based. Use B+ tree indexing, block storage, and LRU caching for acceleration.

2018 — **MASM Assembler**, translating assembly language into machine code
MASM based. Practice knowledge of compiler and linker.

Run-Catch Game, a light-weighted 3D real-time battle game on *WeChat*
Layabox as game engine. Construct 3D models and scenes. Support online real-time playing.

Projects (continued)

- **Contest Platform**, an online system to hold contests for college students
  Django and Vue.js based. Design user-friendly interface, mechanisms to support high concurrency.
- **FTP server and client**, implementing File Transfer Protocol
  Socket based. Implement RFC standards, and the programs interact well with commercial FTP server and client.

2017
- **Object Classification**
  Tensorflow based.
- **XV6 GUI**, adding graphical interfaces to XV6 OS
  Understand the principles of modern OS and details of pixel rendering.
- **Gwent: The Witcher Card Game**, a self-made version of the *game*
  QT based. Complicated game logic, program design, and graphical interfaces.
- **Memory Leak Detector**, a C++ library to discover memory leak
  Check whether a *new* expression is followed by corresponding *delete*.

2016
- **My War Game**, a self-made version of the 2D game *worms reloaded*
  C based. Double buffer rendering and intricate game logics.